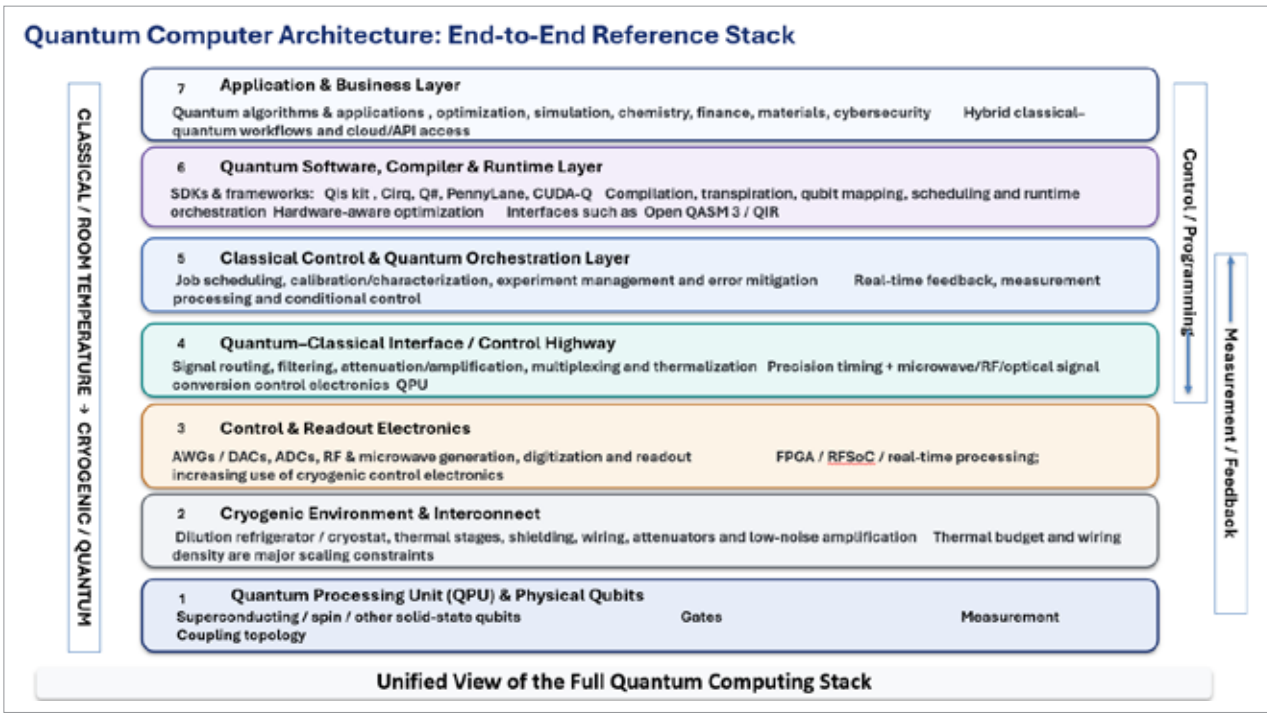


Figure 1: QCP architecture — from cryogenic hardware to business applications

- *Scientific computing:* Physics simulation, quantum chemistry, materials science

Algorithms in quantum computing

A quantum algorithm is a step-by-step procedure, expressed as a quantum circuit, that exploits superposition, entanglement, and interference to solve a problem with fewer operations than the best-known classical algorithm.

Table 2 summarises the algorithms that matter most in 2026.

The current quantum landscape

Various tech leaders are working on quantum computing at present. IBM, Microsoft, Alibaba, D-Wave Systems, Google, Nokia, Fujitsu, and Intel continue to compete for authority in this field.

IBM: IBM has been working to develop a quantum computer for over 35 years. In December 2023, it unveiled a 1,121-qubit processor called Condor, setting a new industry benchmark for qubit count. IBM Quantum Experience is a cloud platform where experiments can be carried out on quantum hardware. Multinational companies such as JPMorgan Chase, Daimler (logistics for optimising vehicle delivery routes), Honda, Samsung and Barclays (settlement of large batches of transactions) were the first to sign up for the active testing of this quantum cloud service.

Google: Google's most recent quantum processor is called Willow. Unveiled in December 2024, Willow is a 105-qubit superconducting chip that for the first time demonstrates exponential error suppression as qubit count scales. Google is aiming for a useful, error-corrected quantum computer by 2029, which is ambitious.

Microsoft: Microsoft has announced Majorana 1, the first QPU to be powered by topological qubits. Azure Quantum provides access to various quantum hardware and quantum-inspired services. Microsoft is also collaborating with other quantum hardware providers to broaden quantum access and accelerate development.

D-Wave Systems: D-Wave built the world's first commercially available quantum computer with a simpler technology but limited in capability. The company is focusing on how quantum computing can be applied to artificial intelligence, machine learning, and difficult optimisation problems. In 2025, it developed a 4400+ qubit called D-Wave Advantage2 to address cybersecurity and logistics optimisation.

Rigetti Computing: Rigetti recently launched a 36-qubit multi-chip processor, which provides a full-stack solution that tightly integrates hardware and software, offering low-latency cloud access for hybrid quantum-classical workflows.

Intel: Intel has announced the delivery of a 49-qubit test quantum processor called 'Tangle Lake', and 'Tunnel Falls', a 12-qubit silicon chip.

Amazon Web Services: AWS introduced its Ocelot chip in February 2025, built around 'cat qubits' designed to reduce error-correction overhead. It continues to operate Braket, a multi-vendor cloud quantum computing service spanning IonQ,